

The slope formula



- 1 Determine the slope of the line that passes through each of the following pairs of points, using $m = \frac{\text{rise}}{\text{run}}$ or $m = \frac{y_2 - y_1}{x_2 - x_1}$.

- | | |
|---|---|
| a Point $A(-1, 0)$ and Point $B(0, 3)$ | b Point $A(2, -6)$ and the origin |
| c Point $A(4, 1)$ and Point $B(11, -6)$ | d Point $A(-3, -1)$ and Point $B(-5, 1)$ |
| e Point $A(0, -4)$ and Point $B(5, -1)$ | f Point $A(-7, -6)$ and Point $B(-5, -8)$ |
| g Point $A(-3, 4)$ and Point $B(1, 4)$ | h Point $A(3, 4)$ and Point $B(5, 4)$ |

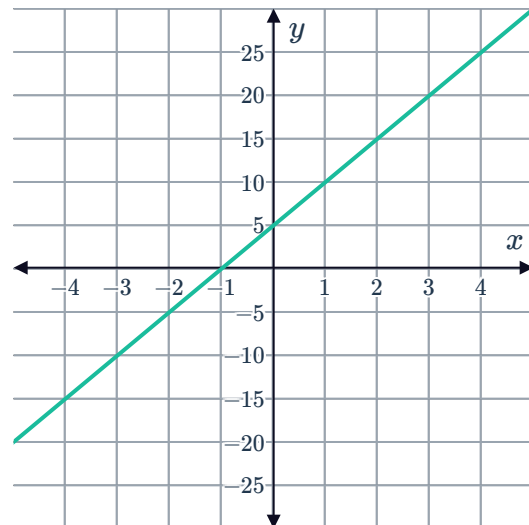
- 2 If we have two points and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2) ? Explain.

- 3 Determine whether the each of the following pairs of coordinates will have a slope that is defined or undefined.

- | | |
|------------------------------|------------------------------|
| a $(-10, 5)$ and $(-10, 12)$ | b $(10, 5)$ and $(10, 1)$ |
| c $(10, -1)$ and $(-10, -1)$ | d $(-10, 5)$ and $(10, 5)$ |
| e $(10, 7)$ and $(10, 2)$ | f $(10, -2)$ and $(-10, -2)$ |
| g $(-10, 7)$ and $(-10, 12)$ | h $(-10, 7)$ and $(10, 7)$ |

- 4 Consider the line $y = 5x + 5$ that has been graphed on the number plane.

- a Find the y value of the point on the line where $x = 4$.
- b Using $m = \frac{\text{rise}}{\text{run}}$, the point found in the previous part and the y -intercept, find the slope of the line.



5 Given the slope of the line passing through two points, find the value of the variable:



a $(4, -3)$ and $(1, t)$, slope $= -2$

b $(5, 3)$ and $(2, t)$, slope $= -4$

c $(5, 3)$ and $(d, 63)$, slope $= 4$

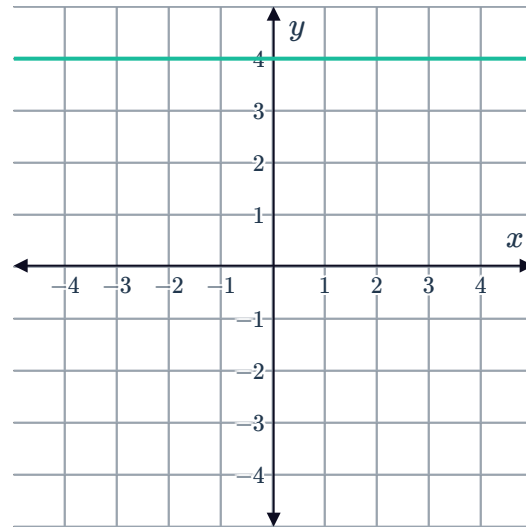
d $(11, c)$ and $(-20, 16)$, slope $= -\frac{4}{7}$

e $(1, 2)$ and $(-4, t)$, slope $= 5$

6 A paratrooper falls to the ground along a diagonal line. His fall begins 1157 m above the ground, and the line he follows has a slope of 1.3. That is, he falls 1.3 m vertically for every 1 m he moves across horizontally. How far horizontally across the ground does he land from his initial position in the sky?

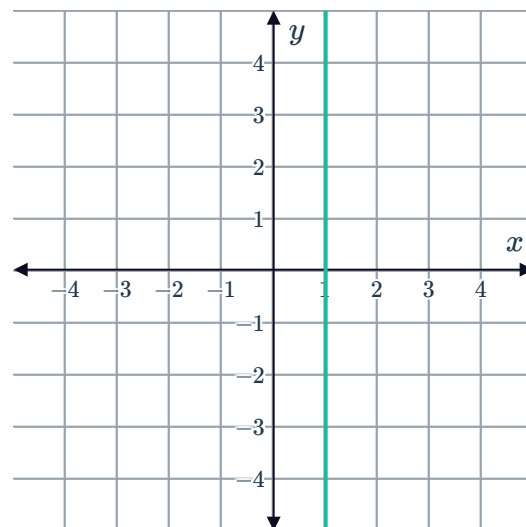
7 Consider the following graph:

- a Determine the slope of the line.
- b Find the y -value of the y -intercept of the line.
- c Does this line have an x -intercept?



8 Consider the following graph:

- a Determine the slope of the line.
- b Does this line have a y -intercept?
- c Identify the x -intercept of the line.

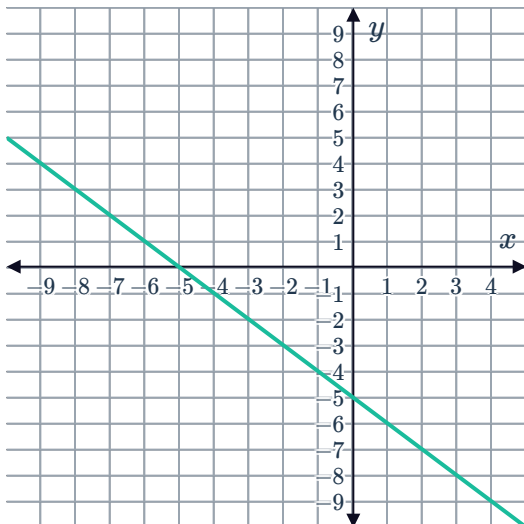


9 Consider the following graphs.

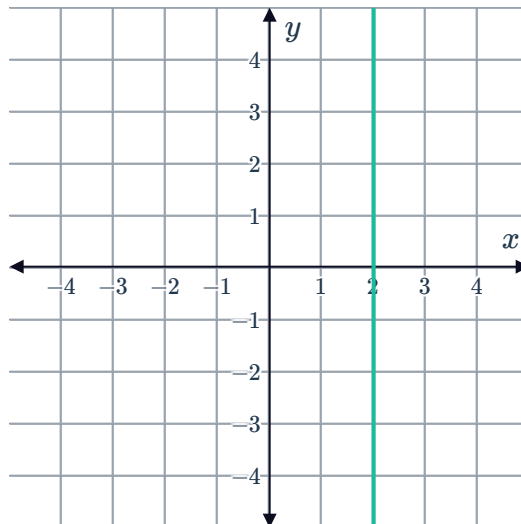


Determine whether the the slope is 0, undefined or neither.

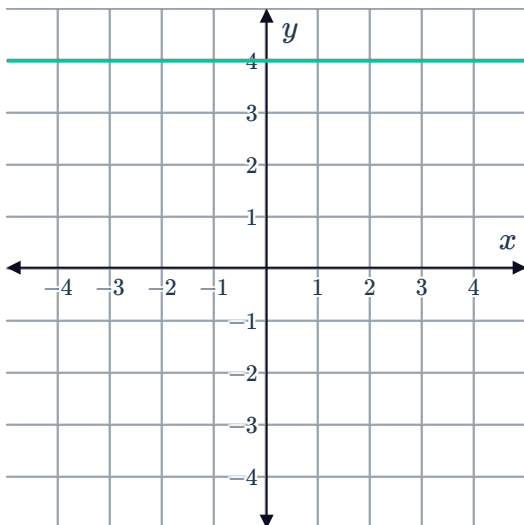
a



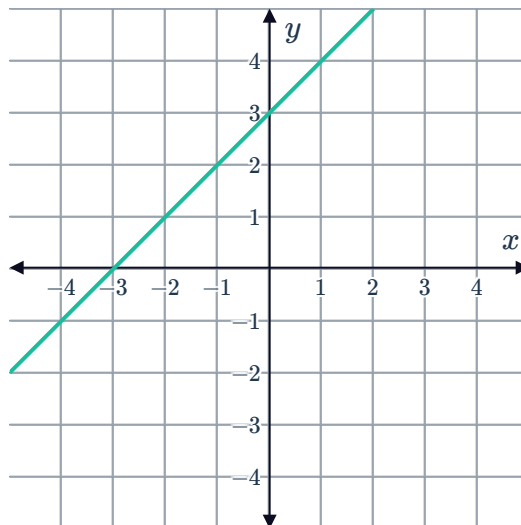
b



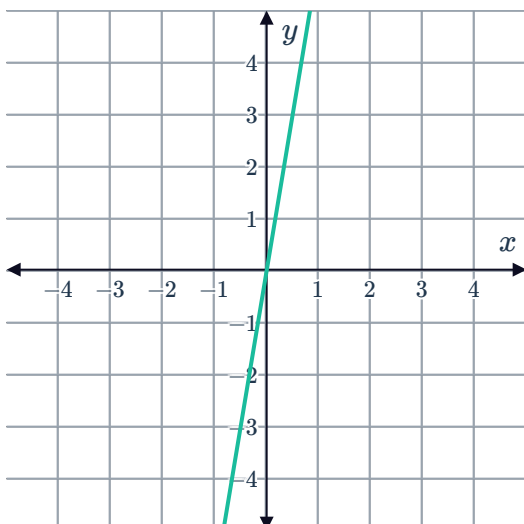
c



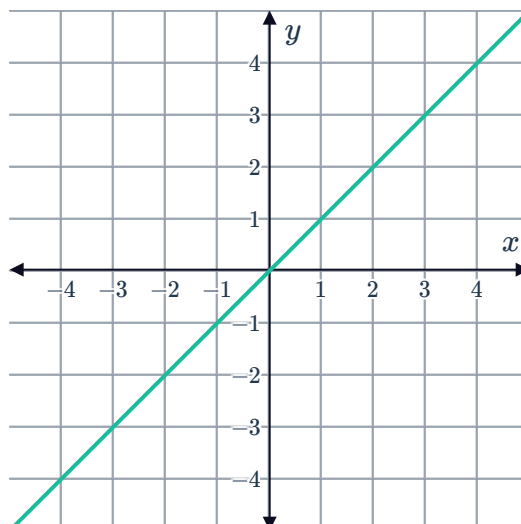
d



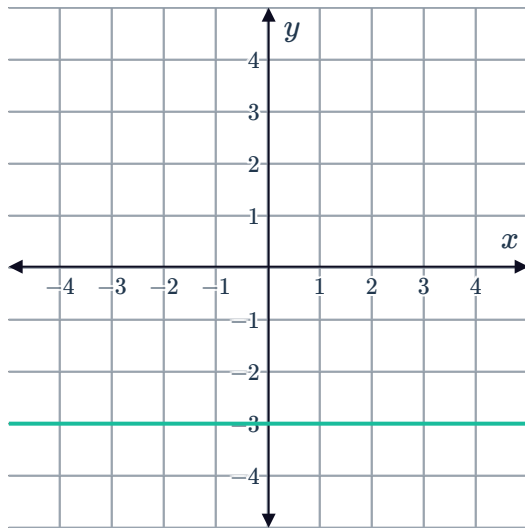
e



f



g



10 Explain what it means if the slope of a line is:

a undefined

b zero

11 State whether each of the following statements is correct for vertical lines.

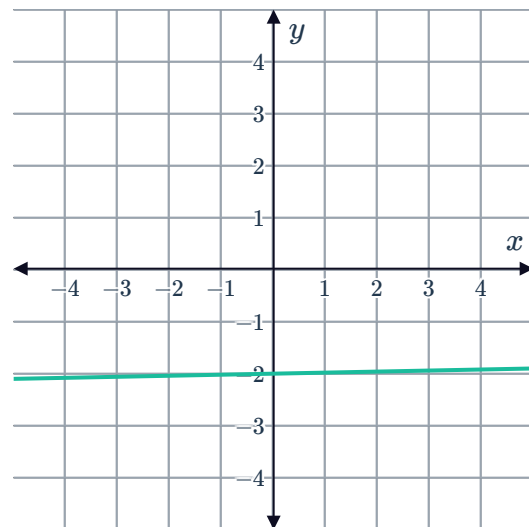
a A vertical line's slope is very large because it is so steep.

b A vertical line has a slope that is undefined because, when calculating the slope using the formula $\frac{\text{rise}}{\text{run}}$, the run is 0 and it is not possible to divide a number by 0.

c A vertical line's slope is equal to 0 because when calculating the slope using the formula $\frac{\text{rise}}{\text{run}}$, the run is 0.

12 The line in the graph is nearly horizontal. Which of the following is most likely to be its slope?

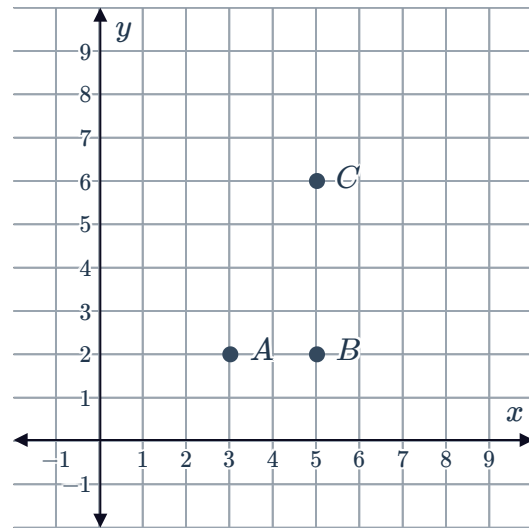
A 0.33 B 0.03 C 3 D 0.3



13 Consider the points on the graph.

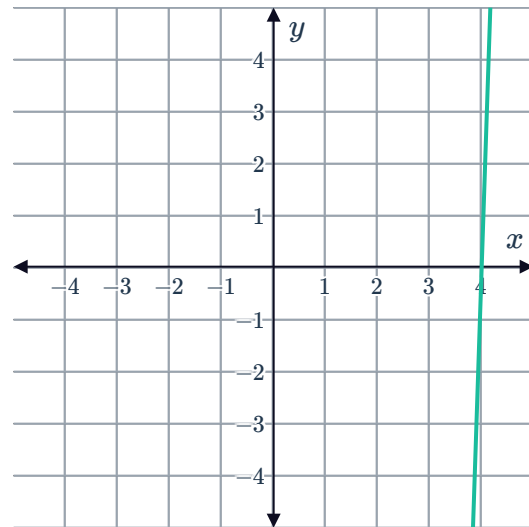


- a Determine the slope of $\overline{AB}$.
- b Determine the slope of $\overline{BC}$.



14 The line that has been graphed is nearly vertical.
Which of the following is most likely to be its slope?

- A** 5 **B** 55 **C** undefined **D** 0.55



15 The function $f(t) = -25t^2 + 30t + 5$ gives the height of a ball, in feet, t seconds after it is thrown.
Determine the average rate of change, in feet per second, over the interval $(0.75, 1.25)$.



Additional questions

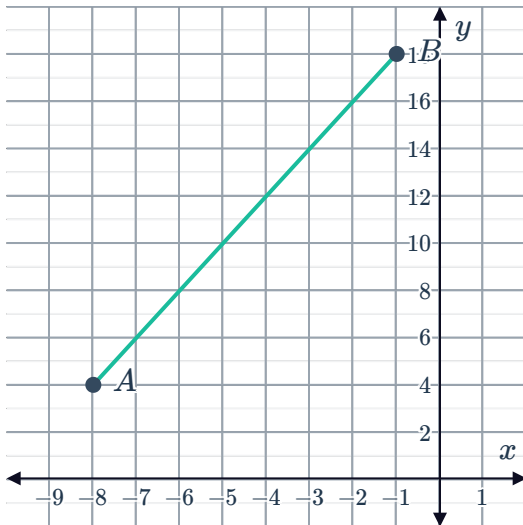
16 For each of the following intervals between points A and B :

i Find the rise.

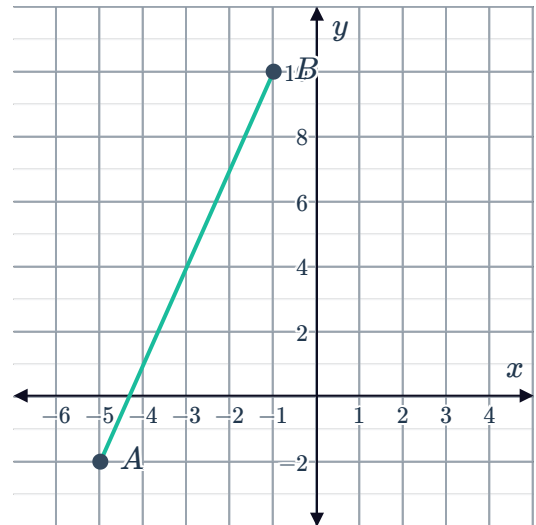
ii Find the run.

iii Find the slope.

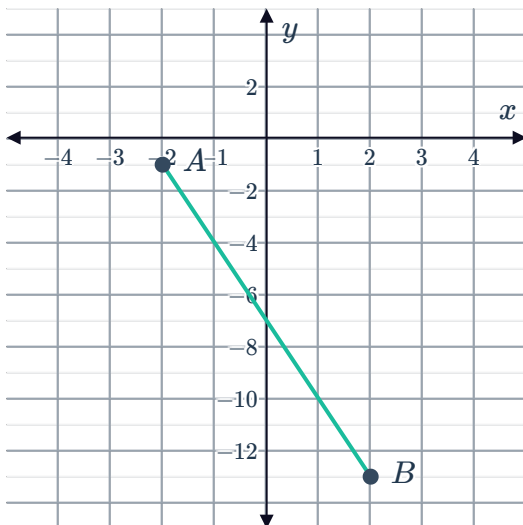
a $A(-8, 4)$ and $B(-1, 18)$



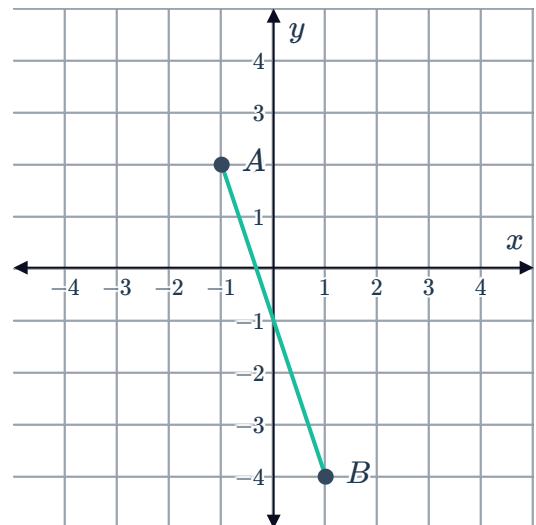
b $A(-5, -2)$ and $B(-1, 10)$



c $A(-2, -1)$ and $B(2, -13)$



d $A(-1, 2)$ and $B(1, -4)$

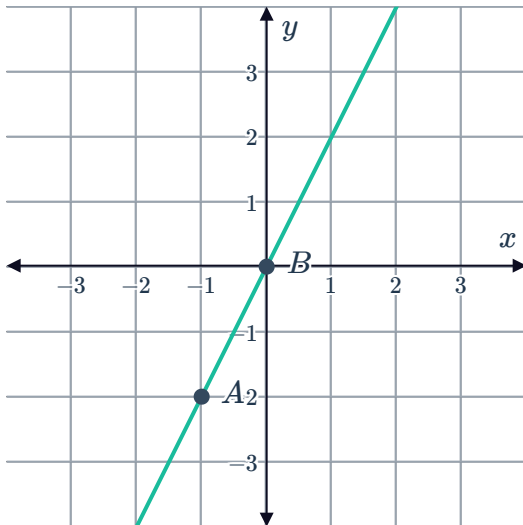


17 Explain why a vertical line has an undefined slope.

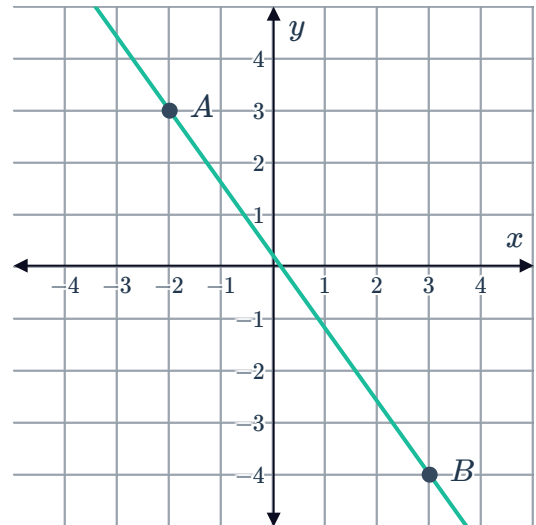
18 State the slope of any line parallel to the x -axis.

19 Determine the slope of the following lines passing through points A and B :

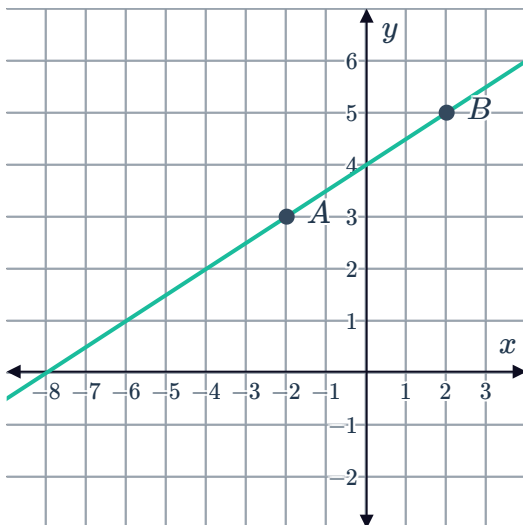
a



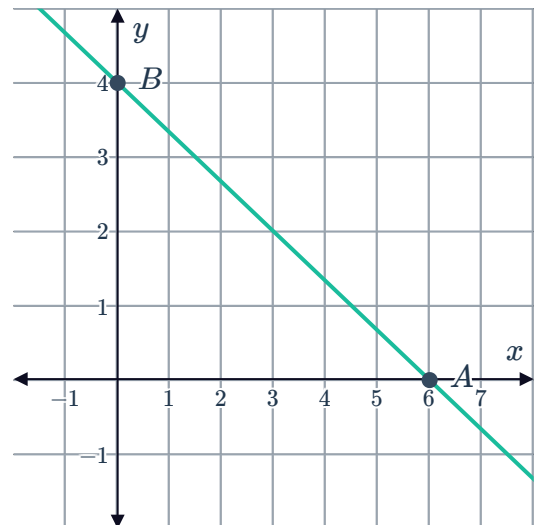
b



c

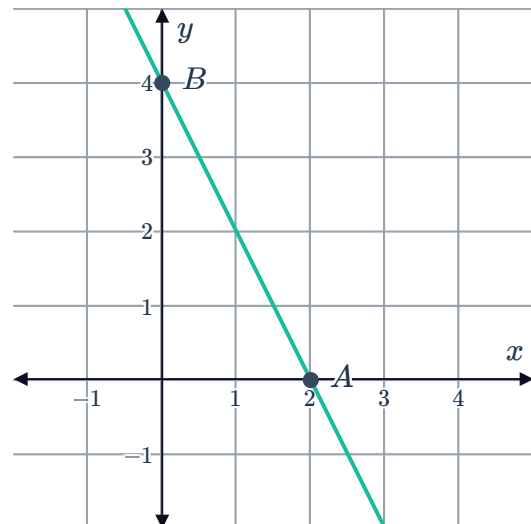


d



20 Consider the line plotted, where $A(2, 0)$ and $B(0, 4)$ both lie on the line.

- a Determine the slope of the line.
- b As x increases, what happens to the value of y ?





21 Determine the slope of the line that passes through the given points:

- | | |
|-------------------------------------|-----------------------------------|
| a $(-1, 0)$ and $(0, 3)$ | b $(-4, 7)$ and $(1, 10)$ |
| c $(1, -4)$ and the origin | d $(2, -6)$ and the origin |
| e $(6, 4)$ and $(3, 4)$ | f $(-6, 5)$ and $(4, 5)$ |
| g $(-2, -5)$ and $(-9, -12)$ | h $(-3, -1)$ and $(-5, 1)$ |

22 Consider the points $A(-11, -9)$, $B(-5, 1)$ and $C(-2, 6)$:

- a** Determine the slope of AB .
- b** Determine the slope of BC .
- c** Do the points A , B and C lie in a straight line?

23 Consider the following points: $A(26, m - 24)$, $B(-1, m)$ and $C(-10, 9)$.
Find m , given that A , B and C are collinear.